Supplement 4: Exploratory Domain Analysis and a Withdrawn Statistic

Joseph Corneli

Oxford Brookes University, Oxford OX3 0BP, United Kingdom

Hyperreal Enterprises Ltd, United Kingdom

Corresponding author: Joseph Corneli, jcorneli@brookes.ac.uk.

Abstract

We study a two-layer cellular automaton in which each cell carries both a phenotype state and an eight-bit rule. A permutation σ of the eight displayed truth-table positions defines an operator P_σ on those rules by $\mathrm{bit}[\sigma(p)] := \neg\,\mathrm{bit}[p]$. The resulting family contains $8! = 40{,}320$ permutation-propagators and is exhaustively classifiable.

The exact, machine-checked classification is that P_σ has a fixed rule exactly when every cycle of σ is even; an all-even permutation with k cycles has 2^k fixed rules; and every fixed rule is balanced, with Langton activity $\lambda = 1/2$. In a census of the spatially coupled dynamics, every odd-cycle operator is genotype-alive under the stated finite-run protocol, whereas every all-even cycle type contains both sustaining and collapsing operators. Within the wrap-rotation subfamily, coprime rotations sustain high diversity while the other nontrivial rotations collapse in the runs tested. A finite-size scan finds a broad, drifting crossover rather than evidence of a sharpening critical point. Filamentary boundaries show elevated genotype rewriting and, in one sampled coexisting-domain field, separate genotypically distinct rule populations at matched activity. A directed-transport statistic on those boundaries does not survive its own controls—mask and measure are derived from the same thresholded field—and is withdrawn here, together with the analysis that rested on it. In its place a causal measure, perturbing one cell and tracking the divergence against a matched control that differs only in the phenotype-to-genotype edge, shows that this feedback roughly doubles how far a perturbation travels, and places the construction between elementary rules 90 and 54 on a damage-spreading scale calibrated against rules of known regime. Temporal braids and a phenotype-reading construction provide two reproducible ways to sustain structure beyond the bare operator.

Keywords: cellular automata, rule evolution, Langton's lambda, self-modifying systems, edge of chaos, artificial life, permutation operators, exhaustive classification, domain walls

This second supplement carries three strands of the companion paper that are reported in full here rather than in the main text: the spatial domain structure of the genotype field and its filamentary boundaries; a measure of information transport along those boundaries that we built, validated against three nulls, and then withdrew; and the domain/particle reading that motivated both.

The main paper's claim is about a single edge of chaos, reached by causal measurement rather than by tuning a parameter. The material here is not that. It is a real spatial structure that we spent some time mistaking for a third edge, and the record of how that mistake was made and caught. We report it at length because the failed construction is a natural one to reach for, and because the geometry-preserving nulls that appear to guard against its failure do not.

Figure, section and empirical-finding numbers below that are not preceded by "Supplement" refer to the main paper, and resolve to its numbering.

Domain Structure and Filamentary Boundaries

The finite-size scan of the main paper (*Supplement 1, Box 6*, Figure 10) finds a broad crossover rather than a critical point. We then asked whether the sustained diversity is spatially organised. It is.

Empirical finding 1: A finite-width intermediate-activity band

An intermediate-activity band ($0.15 \leq a_G \leq 0.60$) has a *lower* edge robustly at $q \approx 0.1$—stable under both system size and threshold—while its upper edge is threshold-dependent, because a_G saturates gradually rather than jumping (the band spans $q \approx 0.1$ to 0.4–0.5). This finite-width crossover is what we call a *littoral edge*: a descriptive name for a shore of finite width, not a thermodynamic phase boundary.

Definition 1 (Isolated-rule activity score)**.** *The* activity score *of a rule is the mean per-step cell-change fraction over the final* 150 *steps of a* 300*-step ordinary ECA run on a periodic line of width* 300*. Every rule uses the same density-*1/2 *pseudorandom initial* *condition (seed* 1*). This finite-run measure is distinct from the static truth-table balance* λ *(*Every Fixed Point Is Balanced*). Ordered rules score near* 0*; Rule* 110 *scores approximately* 0.42*, and several standard chaotic rules score approximately* 0.50*. The score orders activity levels but does not by itself distinguish complex from chaotic dynamics.*

Tinting the genotype by this score (Figure S4.1) resolves the intermediate field into coexisting low- and high-activity domains whose boundaries trace filamentary sets through 1+1-dimensional spacetime—objects *analogous to* the domain boundaries of computational mechanics (Crutchfield & Young, 1989; Shalizi & Crutchfield, 2001), here obtained by thresholding a scalar tint rather than by predictive equivalence.

Empirical finding 2: Filamentary walls have elevated genotype churn

The activity field resolves into ordered and chaotic domains separated by *fractal-like walls* (Figure S4.2; box-counting dimension $D \approx 1.5$–1.7 over $L = 128$–768, a handful of seeds). Those walls are the locus of elevated *genotypic* change: measuring the row-to-row rewrite rate by region (Table S4.1), the wall churn exceeds the chaotic interior in all 32 committed operator/width/seed fields and exceeds *both* interiors in 24. At $L = 256$, offset +1 gives 0.80 at the wall against 0.81 (ordered) and 0.73 (chaotic); the corresponding wall is the maximum region for $\sigma = 16250374$ and the river. The chaotic domain has the lowest aggregate churn for every operator and width (and in 29 of the 32 individual fields), so the wall—where ordered abuts chaotic—is consistently more rewritten than the chaotic interior, though not universally more than the ordered one. The walls are not compositionally special (no rule class enriched; Rule 110 is, if anything, depleted), so the distinction is dynamical rather than a difference in rule composition. The interface analysis is exploratory: it uses a handful of seeds and a fixed-boundary variant, and the denser river boundary ($D \approx 1.8$ versus 1.5–1.7) is an unmatched comparison that does not establish an effect of genotype–phenotype coupling. Whether rewriting transports information *along* the walls is taken up in *A Measure That*

Table S4.1

Genotype churn by region*—row-to-row rewrite rate of the rule field (fraction of cells whose ECA rule changes from the row above), at $L = 256$, mean over the committed seeds. The wall exceeds the chaotic interior for every operator; for offset +1, the ordered interior is slightly higher. Wall extraction is the threshold-and-smooth procedure of Figure S4.2; reproducible from the generated fields (`plot_eoc_churn.py`).*

Operator	Wall	Ordered interior	Chaotic interior
offset +1	0.80	0.81	0.73
$\sigma = 16250374$	0.71	0.68	0.57
river	0.99	0.97	0.96

Failed; the answer there is negative.

The partition is by activity, so the two sides differ in activity by construction. The question that is not self-answering is whether they differ in anything else.

Empirical finding 3: The domains hold different rules at matched activity

Restricting to cells whose *individual* activity score lies in a narrow band, and comparing the distribution over rule identities on the two sides, the total-variation distance exceeds a noise floor—obtained by randomly halving the larger group—by 13 to 25× across bands: 0.489 against 0.023 for scores near zero, 0.147 against 0.011 in $[0.20, 0.35)$, 0.295 against 0.020 in $[0.45, 0.55)$, and 0.522 against 0.021 above 0.55. Rule diversity differs as well: at matched activity in $[0.45, 0.55)$ the active domain holds 24 distinct rules per thousand cells against the quiet domain's 38. Two cells of equal activity belong to systematically different rule populations according to which domain they occupy, so the partition captures genotypic structure the activity score does not determine. Measured on the $q = 0.25$ coexisting-domain field of Figure 10.

A Measure That Failed

If the filaments are domain boundaries they are candidate *wires*. The natural question is not which rules compose them—in our data the complex rules are not enriched there—but whether information is *transported along* them and *combined where they collide* (Mitchell et al., 1993). We built an estimator for both, and both are withdrawn. This section reports the attempt, the controls that appeared to validate it, and the control that defeated it. We keep it in the main text because the construction is one a reader might reasonably build, and because the nulls that seem to guard against its failure do not.

We read both notions off the binary activity field as local transfer entropy (Lizier et al., 2008): the apparent transfer entropy from a spatial neighbour into a cell, conditioned on the cell's own past, scores directed *transport*; the *synergy* of the two neighbours—their joint predictive contribution beyond the sum of their separate ones—scores *combining* (Figure S4.3).

A raw on-filament average will not support either claim, and we say why before using one. The score has two failure modes, and each needs its own null. First, the plug-in estimator is positively biased, so a field can score positive transfer entropy with no directed coupling at all; the null for this is a *source time-shuffle*, which preserves the marginals and the cell's own past and destroys only the neighbour-to-target coupling. Second—and this is what defeats the raw statistic—the extracted filament is not a fixed object across a parameter sweep. It swells from 20% of the field at the bare operator to 39% under strong noise, at which point it is no longer a wall, and any on-filament mean drifts toward the field mean for reasons that have nothing to do with transport. The null for this is a *mask-matched* draw: the same number of cells, taken anywhere in the field. We report the difference from that draw and call it *filament-specific* transport or combining; it is zero when the filament is no better than an arbitrary set of cells of the same size, and negative when it is worse.

A size-matched draw preserves the filament's extent but not its shape, so we also ran two nulls that preserve its geometry. The first circularly translates the mask along x and averages over all displacements beyond the point where a mask ceases to overlap itself; this retains the mask's exact connectivity, anisotropy, cell count and per-row occupancy, destroying only its registration against the field, and it involves no randomness at all. The second scores each field under the filament extracted from an independent seed at the same sweep point. Both agree with the size-matched draw to within 0.001 bits at every point ($r > 0.999$ across the sweep), and every result below is unchanged under all three. A filament-shaped region placed generically therefore scores no differently from scattered cells: what distinguishes the real filament is where it lies, not the shape of the region measured. We quote the size-matched null throughout.

The estimator has a third failure mode that no null removes, and it bounds what follows. Where the activity field is nearly degenerate—almost all cells in one state—the conditional probabilities driving the score become extreme and it inflates without bound, and this happens at *either* end. The genotypically dead offset +4 operator is the demonstration. Its genotype freezes (row-to-row rewrite rate falls to 0, Empirical *Supplement 1, Box 3*) into a population that is 98% active, and on that saturated field it records the largest on-filament transport we measured anywhere: 317× the domain interior, +0.59 bits filament-specific, for an operator that transports nothing at all. The sparse end behaves the same way, and costs us the most ordered sample in the sweep below (all blend, $q = 0$, 9% active), where the shuffled surrogate outscores the real field (+0.144 against +0.105 bits, $t = -0.71$, $p = 0.49$); that point is bias-dominated and is excluded. Local transfer entropy is therefore comparable only across fields of similar and non-degenerate activity density—which is the concrete content of what earlier drafts recorded as a "sparsity effect".

Within these bounds the score behaves as one would hope, and it is worth recording what that looked like, because it is what made the result credible. Over the eight seeds of the $q = 1$ sweep column, on-filament transfer entropy is $+0.158 \pm 0.005$ bits against a shuffled surrogate of +0.002 ($t = 33.5$, $p < 10^{-8}$) and a mask-matched draw of +0.061, leaving $+0.097 \pm 0.005$ bits filament-specific. On the fixed-boundary field of Figure S4.2 the measure concentrates 5.7× over the domain interiors; the Rule 110-dominated $\sigma = 16250374$ field, taken as a positive control, shows the same signature (7.6×, filament-specific +0.161 bits). Three nulls agree to within 0.001 bits, the positive control behaves as a positive control should, and the effect is large and highly significant. None of that is evidence, for the reason given next.

The measure does not survive its own controls

The score defined above fails a control we did not originally run. Local transfer entropy here is computed on the thresholded activity field act = (smooth > 0.35), while the filament mask is the *gradient of that same field*. The mask therefore marks precisely the cells at which act changes, and transfer entropy on a binary field is near zero except where it changes: +0.8934 bits on changed cells against +0.0080 elsewhere, the wall capturing changed cells at 163× enrichment. A positive filament-specific score is guaranteed by the construction rather than by the dynamics.

Two tests establish this. Holding the mask at 0.35 and recomputing the measure at a different threshold collapses the score from +0.0885 to +0.0007 (at 0.30), −0.0044 (0.40) and −0.0377 (0.50). Deriving mask and measure from the same threshold and sweeping it instead recovers the effect at *every* threshold, growing monotonically from +0.065 at 0.25 to +0.173 at 0.50. A physical quantity should not track an arbitrary analysis parameter monotonically.

The three nulls do not defend against this and could not have. Size-matched, shift and cross-seed each control a property of the *mask*—its cell count, its shape, its provenance—and every one of them displaces the mask off the change-cells. All three therefore report a large effect *because* of the circularity rather than despite it. This is a gap in the null design, not a fault in the nulls as implemented.

Nor is it ordinary sensitivity to how one draws a region. Four independent drawings, scored on the same fields with size-matched masks, order themselves by how much construction each shares with the measure: the thresholded gradient gives +0.0885, the gradient of the *continuous* activity field +0.0200, local causal states +0.0045, and a rule-identity boundary +0.0007. Genuine algorithm-dependence would scatter; this ranks.

Replacing the measure while keeping the mask also returns nothing. On the phenotype field, which shares no threshold with the mask, the filament-specific score is negative on all six seeds (mean −0.011) and −0.160 on the two-4-cycle field.

We therefore withdraw both claims—that these boundaries carry directed transport, and that they combine information where they meet. Combining is the synergy of the same two neighbours read off the same thresholded field under the same mask, so it inherits the circularity intact, and it likewise fails the decoupled test (phenotype-field combining is +0.003 against the size-matched null, against +0.028 for the activity field). With them goes the sweep that rested on the same statistic. What survives is reported in *A Parameter-Space Test*: the walls are where genotype rewriting is fastest, and they separate genotypically distinct populations. Those rest on different measurements—local transfer entropy is uncorrelated with genotype churn ($r \approx 0.02$)—and do not inherit this failure.

Domains and Particles

The structures reported above have an ancestor. Computational mechanics reads a cellular-automaton spacetime diagram not as one pattern but as a decomposition: *domains*, the regular backgrounds a rule settles into; *particles*, the localised structures that separate one domain from another and propagate through the diagram; and *collisions*, the interactions between particles, where the system's computation is usually held to be taking place (Crutchfield & Hanson, 1993; Hanson & Crutchfield, 1997). Rule 110's gliders are the canonical case. *The Object* places this paper's *object* among cellular-automaton variants; this section places its *structures* among ways of reading them.

The correspondence runs at all three levels. Tinting the genotype by activity resolves the intermediate field into coexisting low- and high-activity regions (Figure S4.1):

domains, in the sense above, though not obtained in the same way. The filamentary walls separating them are localised, persistent, and propagate through spacetime with box-counting dimension 1.5–1.7 (Empirical finding 2); they occupy the position particles occupy, and they are where genotype rewriting is fastest. The river (Empirical *Supplement 1, Box 7*) produces coherent bands that drift through a disordered background and then branch, merge, and reorganise—which is what a population of interacting particles looks like. That last resemblance is qualitative and we do not press it: the river's denser boundary is an unmatched comparison and establishes nothing about genotype–phenotype coupling.

Where this departs from the ancestor is the reason the walls move. In the classical setting one fixed rule runs everywhere; a domain is a pattern in the phenotype, and a particle is a defect in that pattern, carried along by the single rule that generates both. Here the domains are regions of *different rules*, and a wall is mobile because the genotype is being rewritten underneath it. Their particles move through a static rule; ours move because the rule itself is being edited at the boundary. The two pictures agree about where to look and disagree about what is doing the moving.

We inherit the vocabulary, not the machinery. Computational mechanics obtains its domains by predictive equivalence, building the ϵ-machine of the spatial configuration and reading regularity off it (Crutchfield & Young, 1989; Shalizi & Crutchfield, 2001); ours come from thresholding a scalar activity tint, which is cheaper, coarser, and carries a caveat sharper than the one usually recorded: a statistic computed on a region drawn from the same thresholded field cannot be trusted, as *A Measure That Failed* shows at length. What the tint supports is the partition itself (Empirical finding 3), not quantities evaluated on its boundary. The mapping above is therefore a reading aid rather than a derivation. A properly computational-mechanical treatment of this substrate—domains defined by predictive equivalence over the joint genotype–phenotype configuration, particles as their boundaries—remains the natural next instrument. A first attempt, retargeting a held-out local causal-state reconstruction at these fields, agrees with the tint partition only at chance ($\kappa = 0.008$ over six seeds) and finds no transport of its own; its depth and tolerance selected at the edge of the search grid, so it is not yet decisive either way.

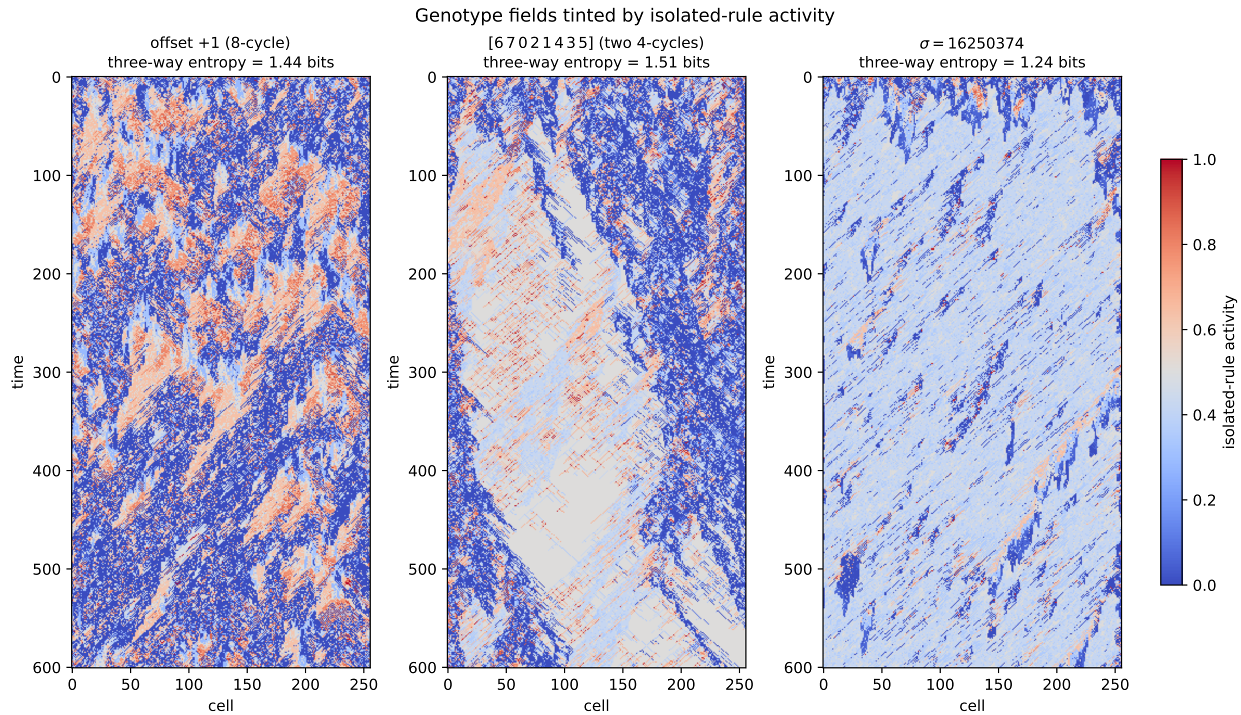

Figure S4.1

The genotype as a map of activity regimes. *Genotype spacetime fields (fixed-boundary variant) tinted by each rule's* isolated-rule activity score *(Definition 1; blue = low near* 0*, pale = intermediate near* 0.4*, red = high near* 0.5 *and above). Which active regime pairs with the low-activity background depends on the operator, and the three-way balance differs sharply (entropy in bits, in title): offset* $+1$ *(the 8-cycle survivor) mixes low with high-activity chaotic rules in a* fine *texture; the two-4-cycle* $[6\,7\,0\,2\,1\,4\,3\,5]$*—the standard-order form of the historical offset*$+2$ *exemplar—forms* large coherent *intermediate-activity domains with sharp walls, the most balanced regime mix (entropy* 1.51*); and* $\sigma = 16250374$ *is dominated by the intermediate complex Rule 110. The score orders the regimes but does not by itself separate complex from chaotic.*

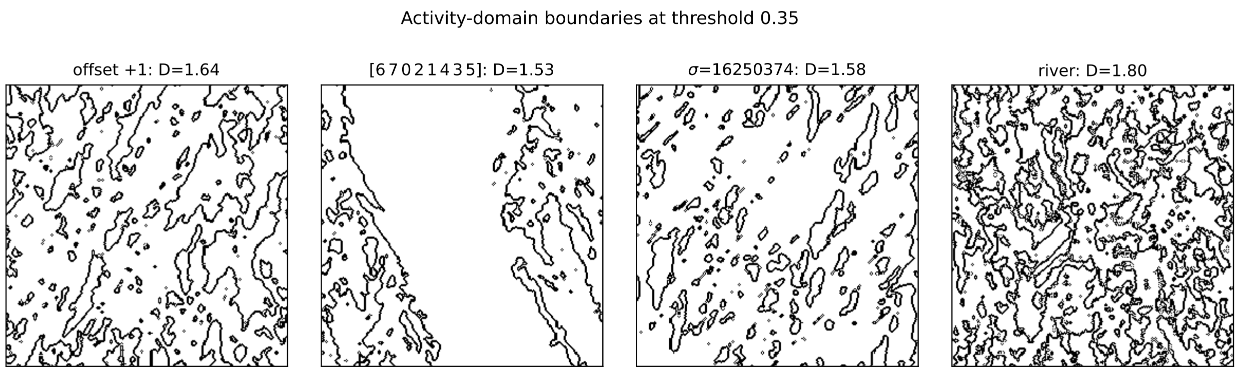

Figure S4.2

Activity-domain walls. *The boundary (black) between low- and high-activity regions, traced through the* 1+1*-dimensional spacetime diagram at* $L = 256$*. Offset* $+1$*, the two-4-cycle* $[6\,7\,0\,2\,1\,4\,3\,5]$*, and* $\sigma = 16250374$ *carry fractal-like boundaries (*$D \approx 1.5$*–*1.6*); the phenotype-coupled river carries a denser one (*$D \approx 1.80$*, an unmatched comparison as it couples phenotype). The dimension is stable across* $L = 128$*–*768 *(offset* $+1 \approx 1.67$*, the two-4-cycle* ≈ 1.50*,* $\sigma \approx 1.54$*, and river* ≈ 1.79*; box sizes* 2*–*64*). Collapsed operators (e.g. offset* $+4$*) carry none. These walls are also consistently more rewritten than the chaotic-domain interior (Discussion). Exploratory: a handful of seeds, fixed-boundary variant.*

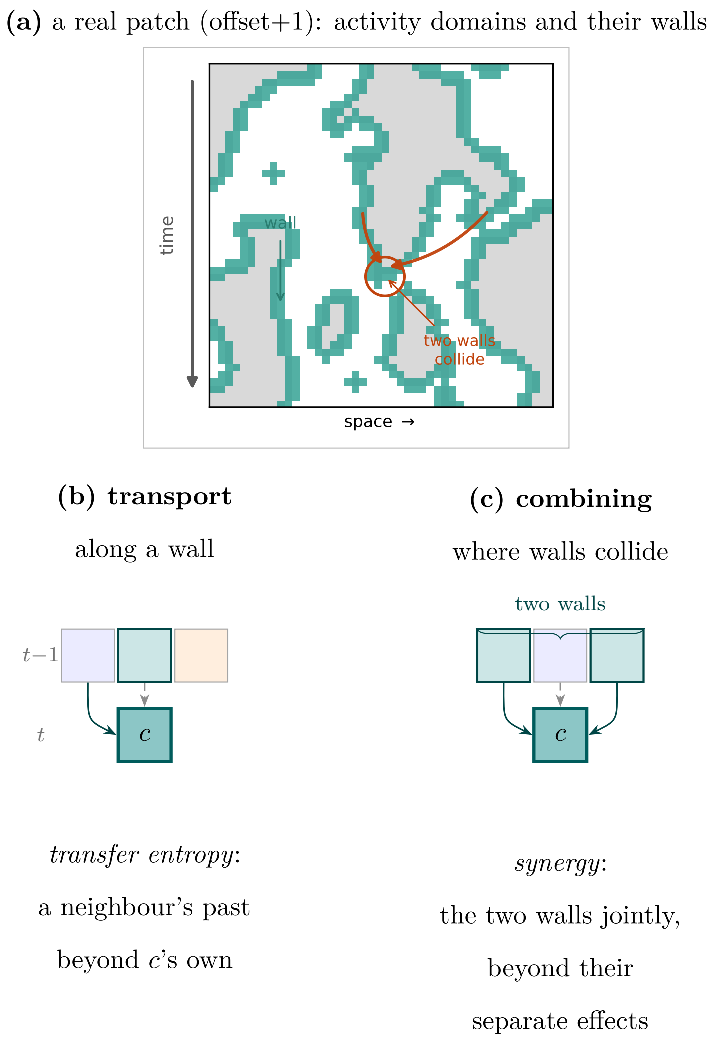

Figure S4.3

How the two (withdrawn) measures read a filament. *The estimator illustrated here is the one this section invalidates; the diagram is retained to make the failure legible.* ***(a)*** *A real space–time patch from the synthetic family (offset+1): low- and high-activity domains (white/grey) meet along thin walls (teal), with one wall and one collision marked. In the schematics each cell c is predicted from its own past (dashed) and its two spatial neighbours' pasts (teal); tints mark the low- (blue) and high-activity (orange) domains a wall separates.* ***(b)*** Transport *is the directed transfer entropy from a neighbour into c beyond c's own past, which the wall is defined to maximise—the circularity of* A Measure That Failed*.* ***(c)*** Combining *is the* synergy *of the two neighbours—their joint contribution beyond the sum of their separate ones—probed where two walls collide. History length one; conditioning on the own past removes what the wall merely carries forward in time.*

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